EDS 223: Homework Assignment 3

Prioritizing Potential Aquaculture Along the West Coast

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Load in Packages and Data

```
# Load in libraries
library(tidyverse)
library(terra)
library(sf)
library(tmap)

# Load in sea surface data
sst_2008 <- rast("data/average_annual_sst_2008.tif")
sst_2009 <- rast("data/average_annual_sst_2009.tif")
sst_2010 <- rast("data/average_annual_sst_2010.tif")
sst_2011 <- rast("data/average_annual_sst_2011.tif")
sst_2012 <- rast("data/average_annual_sst_2012.tif")

# Load in depth data
depth <- rast("data/depth.tif")

# Load in EEZ data
eez <- st_read("data/wc_regions_clean.shp")
```

Stack and Process Data

```
# Combine yearly raster layers into one raster
sst_stack <- c(sst_2008, sst_2009, sst_2010, sst_2011, sst_2012)
```

```r
# find mean SST from 2008-2012 and convert from K to C°
avg_sst_C <- mean(sst_stack, na.rm = TRUE) - 273.15

# Resample the depth data to match resolution of SST
depth_resample <- resample(depth, avg_sst_C, method = "bilinear")
```

Check layers have same resolution, extent, and CRS

```r
# Checking and converting depth to match SST resolution
if (all(res(avg_sst_C) == res(depth_resample))) {
  print("The resolutions match")
} else {
  warning("Update resolutions to match")
  res(depth_resample) <- res(avg_sst_C)
}
```

```
[1] "The resolutions match"
```

```r
# Checking and converting depth to match SST extent
if (ext(avg_sst_C) == ext(depth_resample)) {
  print("The extents match")
} else {
  warning("Update extents to match")
  ext(depth_resample) <- ext(avg_sst_C)
}
```

```
[1] "The extents match"
```

```r
# Checking and converting depth to match SST CRS
if (crs(depth_resample) == crs(avg_sst_C)) {
  print("The CRSs match")
} else {
  warning("Update CRSs to match")
  crs(depth_resample) <- crs(avg_sst_C)
}
```

```r
# Function to make maps of suitable aquaculture area

# Inputs:
#   min_sst, max_sst     - suitable sea surface temperature range (°C)
#   min_depth, max_depth - suitable depth range, as POSITIVE meters below
#   species_name         - species name for organization and the map title
#
# This function requires data prepared from potential_aquaculture.qmd:
#   avg_sst_C      - mean 2008-2012 SST in °C
#   depth_resample - depth cropped and resampled to match avg_sst_C
#   eez            - West Coast eez polygons, ensure the same CRS as rasters

suitable_zones <- function(
  min_sst,
  max_sst,
  min_depth,
  max_depth,
  species_name
) {
  # Reclassify SST by making matrix
  rcl_sst <- matrix(
    c(
      -Inf,
      min_sst,
      0,
      min_sst,
      max_sst,
      1,
      max_sst,
      Inf,
      0
    ),
    ncol = 3,
    byrow = TRUE
  )

  # Apply it to raster layer
  sst_zones <- classify(avg_sst_C, rcl = rcl_sst)

  # Reclassify depth by making matrix
  rcl_depth <- matrix(
    c(
```

```
    -Inf,
    -max_depth,
    0,
    -max_depth,
    -min_depth,
    1,
    -min_depth,
    Inf,
    0
  ),
  ncol = 3,
  byrow = TRUE
)

# Apply it to raster layer
depth_zones <- classify(depth_resample, rcl = rcl_depth)

# Find areas that satisfy both SST and depth
species_stack <- c(sst_zones, depth_zones)

species_suitable <- lapp(species_stack, fun = function(sst, depth) {
  sst * depth
})

# Rastertize EEZ
eez_raster <- rasterize(vect(eez), species_suitable, field = "rgn_id")

# Calculating area of each cell
cell_area <- cellSize(species_suitable, unit = "km", mask = TRUE)

# Keep area of suitable cells only; unsuitable (0) become NA
suitable_area <- mask(cell_area, species_suitable, maskvalues = 0)

# Select raster cells that fall in EEZ regions
suitable_area_eez <- zonal(
  suitable_area,
  eez_raster,
  fun = "sum",
  na.rm = TRUE
)

# Convert into DataFrame
```

```r
suitability_df <- as_tibble(suitable_area_eez) |>
  rename(suitable_area_km2 = area) # rename column name

# Join suitable areas to EEZ polygons
eez_map <- left_join(eez, suitability_df, by = "rgn_id")

# Make a map of West Coast EEZ's
tm_shape(eez_map) +

  # Add a basemap
  tm_basemap("OpenStreetMap") +

  # Customize legend
  tm_fill(
    fill_alpha = 0.75,
    fill = "suitable_area_km2",
    fill.legend = tm_legend(
      "Suitable Area (km²)",
      position = tm_pos_out("right", "center")
    )
  ) +

  # Customize border of polygons
  tm_borders(
    lwd = 0.5
  ) +

  # Add a scalebar
  tm_scalebar(
    position = c("left", "bottom"),
    text.size = 0.25
  ) +

  # Add a compass
  tm_compass(
    position = c("right", "top"),
    size = 1.5
  ) +

  # Add a title
  tm_title(
    paste("Suitable Aquaculture Areas for", species_name, "by EEZ"),
```

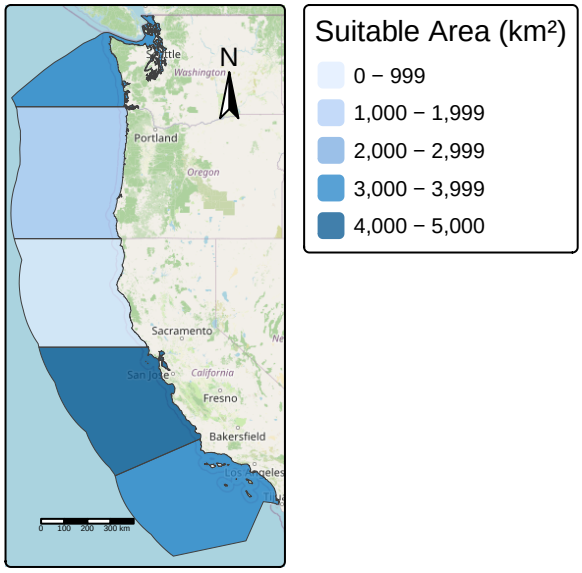
```
      fontface = "bold",
      size = 1
    ) +

    # Remove warning
    tm_options(component.autoscale = FALSE)
}
```

Maps

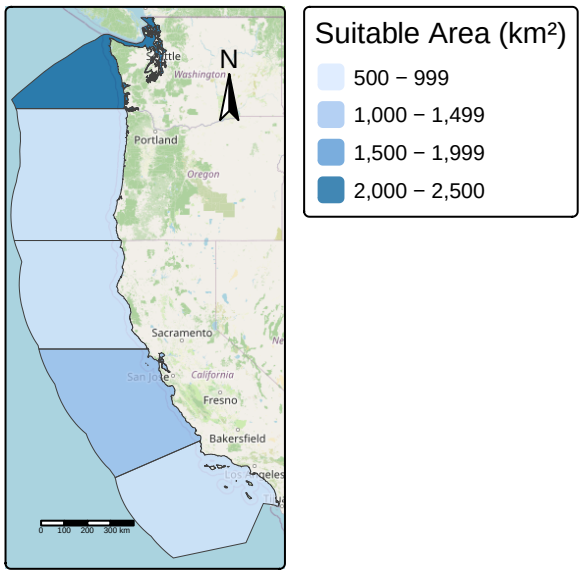
```
# Suitable zone map for oysters
suitable_zones(
  min_sst = 11,
  max_sst = 30,
  min_depth = 0,
  max_depth = 70,
  species_name = "Oysters"
)
```

Suitable Aquaculture Areas for Oysters by EEZ



```
# Suitable zone map for red abalone
suitable_zones(
  min_sst = 8,
  max_sst = 18,
  min_depth = 0,
  max_depth = 24,
  species_name = "Abalone"
)
```

Suitable Aquaculture Areas for Abalone by EEZ



Reflection

The Central California EEZ contains the largest area suitable for oyster aquaculture, followed by the Southern California and Washington EEZs. Oregon has a moderate amount of suitable area, and Northern California has the least. Suitable areas were defined by the optimal growth conditions for oysters which require an average sea surface temperature (SST) between 11°C and 30°C, and a depth of 0 to 70 meters below sea level.

The Washington EEZ contains the largest area suitable for red abalone aquaculture, followed by the Central California EEZ. The Oregon, Northern California, and Southern California EEZs have the least suitable area. Suitable areas were defined by the optimal growth conditions for red abalone which require an average sea surface temperature (SST) between 8°C and 18°C, and a depth of 0 to 24 meters below sea level.

To identify these areas, annual average SST rasters from 2008 to 2012 were stacked and averaged into a single raster, then converted from Kelvin to Celsius. The bathymetry raster was cropped and resampled to match the SST grid so the two layers could be combined. Each layer was reclassified into suitable (1) and unsuitable (0) cells, and multiplying the layers identified cells that met both conditions. Exclusive Economic Zones were used to define maritime boundaries, allowing suitable area to be summed within each zone and compared along the West Coast. This workflow was then generalized into a function that can map suitable area for any species with known SST and depth ranges.

This analysis has several limitations. Suitability was based only on SST and depth and other important factors such as dissolved oxygen, ship traffic, and marine protected areas were not considered. Averaging SST across five years also hides seasonal extremes that could stress oysters even in areas classified as suitable.